

Ans: B

Q25

For potassium, $hc/\lambda = \phi_{pot} + (1/2)mv_{max}^2$

$$hc/(250 \times 10^{-9}) = 2.24 \times 1.6 \times 10^{-19} + (1/2)(9.11 \times 10^{-31})v_{max}^2$$

$$v_{max} = 9.80 \times 10^5 \text{ m s}^{-1}$$

For zinc, $hc/\lambda = \phi_{zn} + (1/2)mv_{max}^2$

$$hc/(250 \times 10^{-9}) = 4.33 \times 1.6 \times 10^{-19} + (1/2)(9.11 \times 10^{-31})v_{max}^2$$

$$v_{max} = 4.75 \times 10^5 \text{ m s}^{-1}$$

$$\text{Ratio of speeds} = 9.80 \times 10^5 / (4.75 \times 10^5) = 2.1$$

Ans: C

Q26